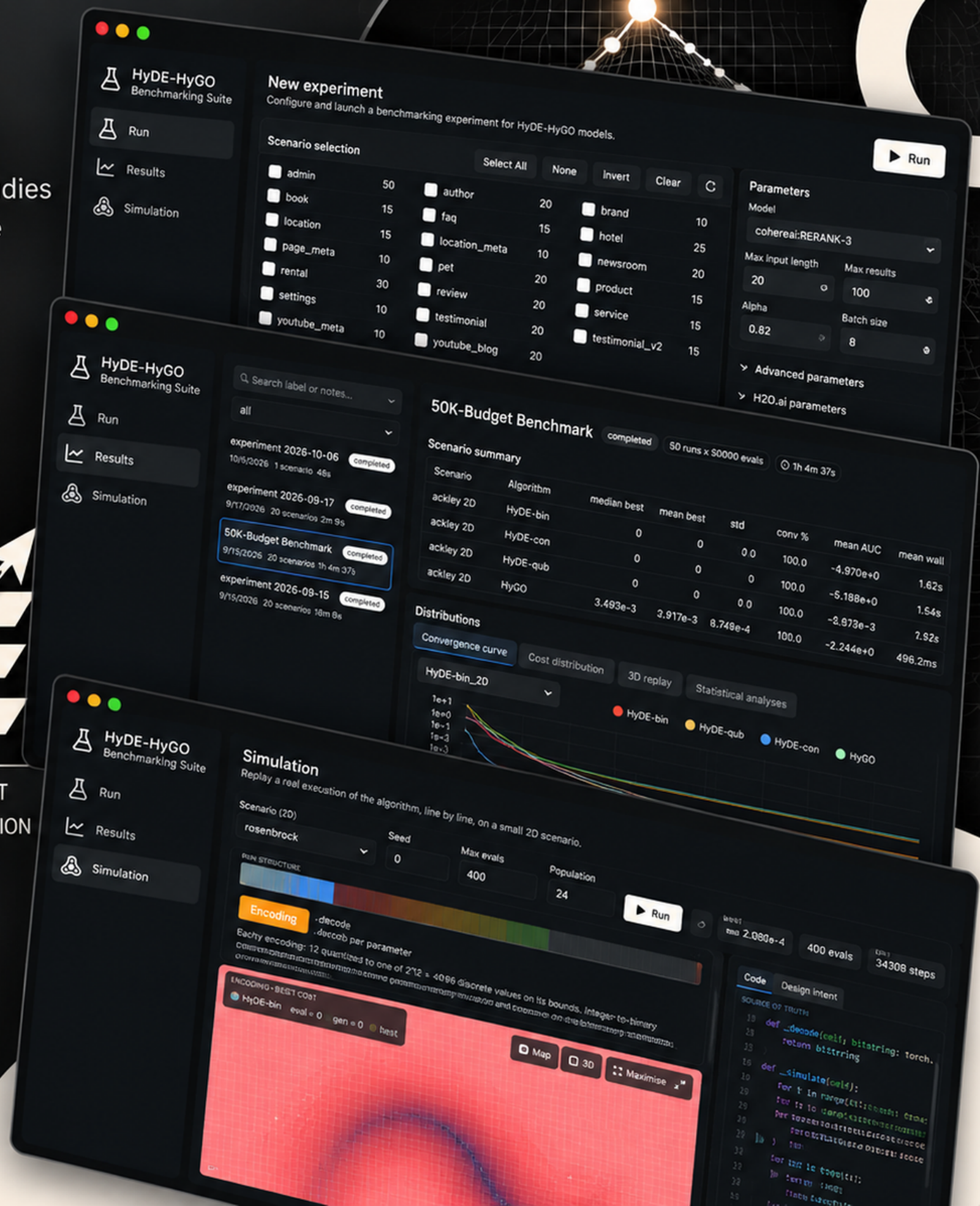
DIVINE WORD COLLEGE OF LEGAZPI

School of Engineering and Computer Studies

Bachelor of Science in Computer Science

HYBRID DIFFERENTIAL EVOLUTION WITH WARM-START INCREASING POPULATION COVARIANCE MATRIX ADAPTATION EVOLUTION STRATEGY FOR CONTINUOUS PARAMETRIC OPTIMIZATION

2026 SIMULATION



SYSTEM PROPERTIES, KEY FEATURES & DEVELOPERS

SYSTEM PROPERTIES

PLATFORM	Desktop benchmarking and simulation application
ALGORITHMS	HyGO, HyDE-BIN, HyDE-QUB, HyDE-CON
CORE COMPUTATION	Python 3.10+; NumPy and SciPy
ANALYSIS & PLOTS	Pingouin, Pandas and Matplotlib
FRONTEND	React TS, PyTauri
BACKEND	Python 3.10, NumPy, SciPy
DEVELOPMENT TOOLS	VS Code, Jupyter Notebook

KEY FEATURES

- Configure and run repeatable benchmark experiments
- Compare four algorithms on analytical benchmark functions
- Inspect convergence, computational cost and statistical analyses
- Visualize optimization search behavior in 2D and 3D

DEVELOPERS

Manuel II M. Caballero

Denzel Jay B. Benjamin

Christian Nickhos A. Divina

ACCESS THE NEW EXPERIMENT SCREEN

STEP-BY-STEP WORKFLOW / PART 1 OF 2

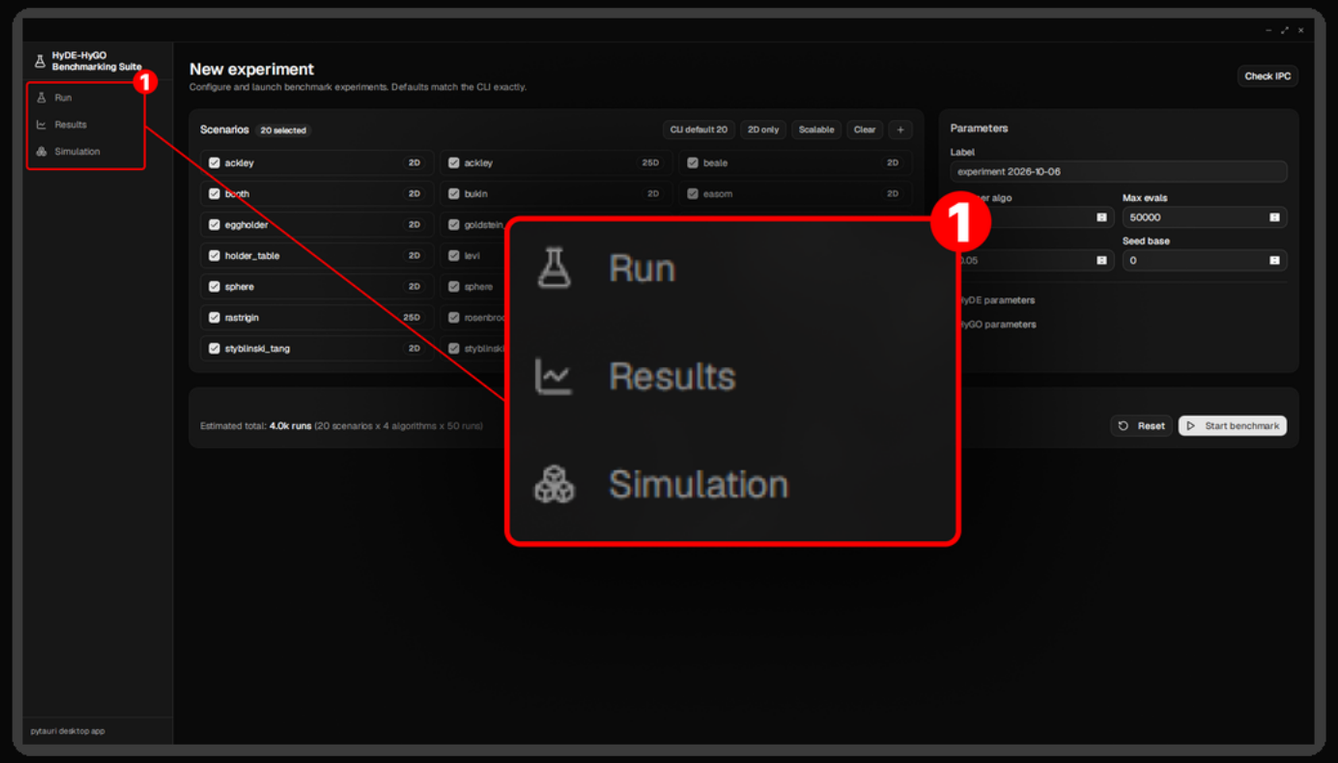
INSTRUCTIONS

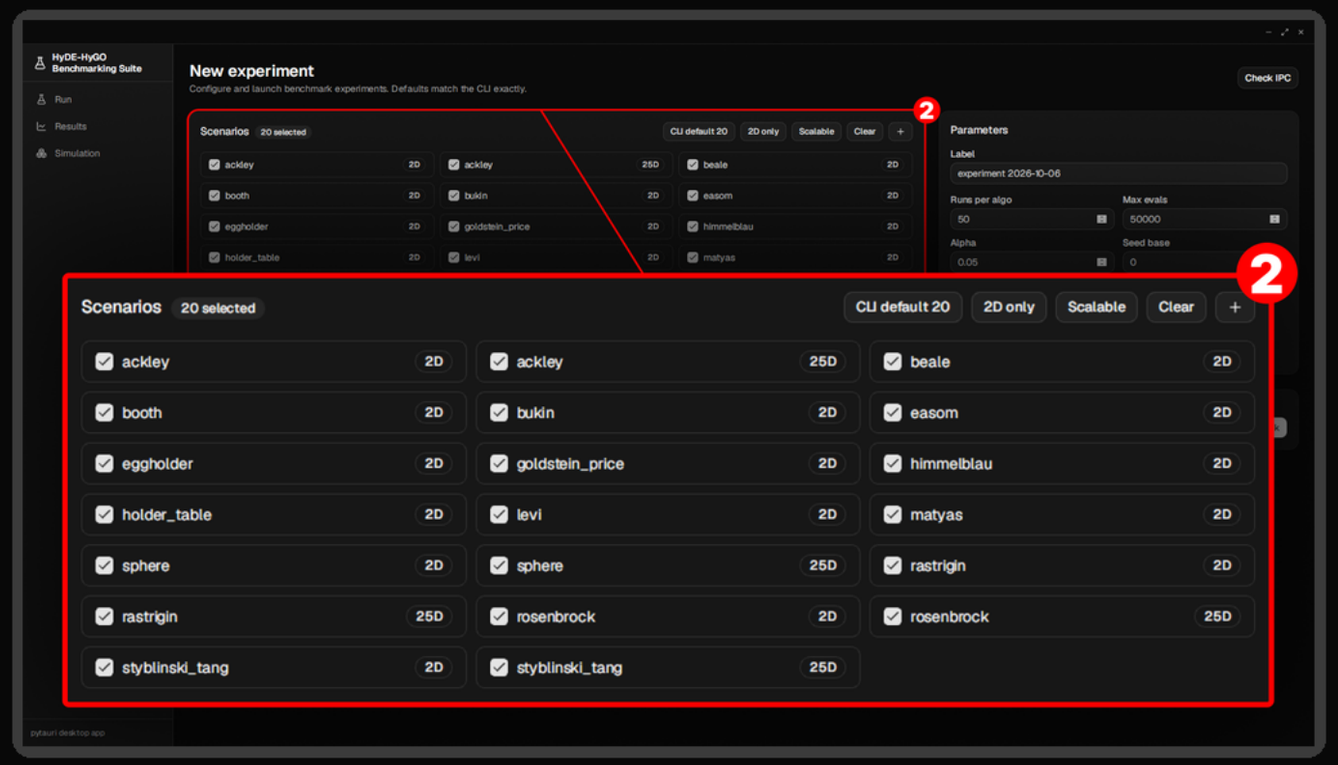
Once users open the BenchSuite, they will be directed by default to the Run Page, where users can:

1. Move between the major application areas using Sidebar Menu items (Run, Results, or Simulation);

2. Select Scenarios checkboxes or use the shortcut buttons (CLI Default 20, 2D Only, Scalable, Clear, and +) to choose which test functions (standard problems for algorithm comparison) will be benchmarked in 2D or 25D;

2 annotated screenshots





ACCESS THE NEW EXPERIMENT SCREEN

STEP-BY-STEP WORKFLOW / PART 2 OF 2

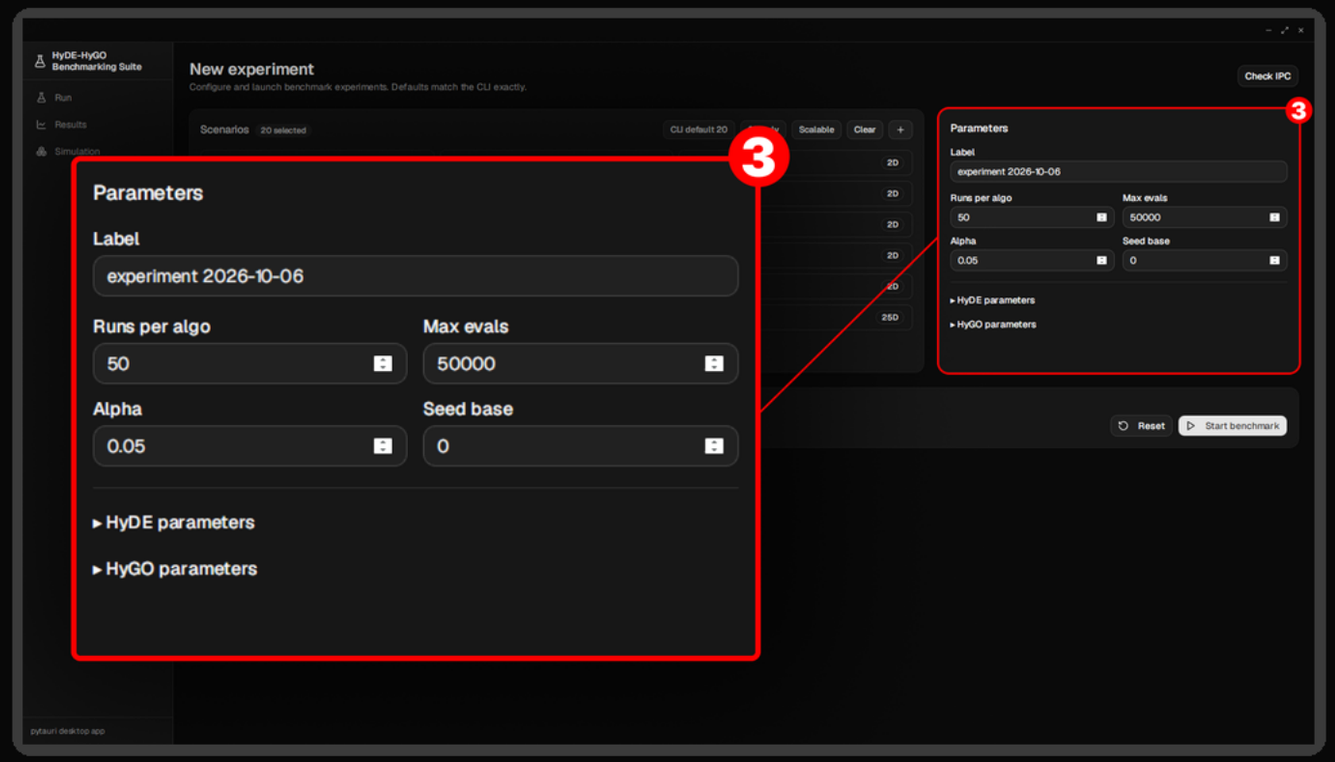
INSTRUCTIONS

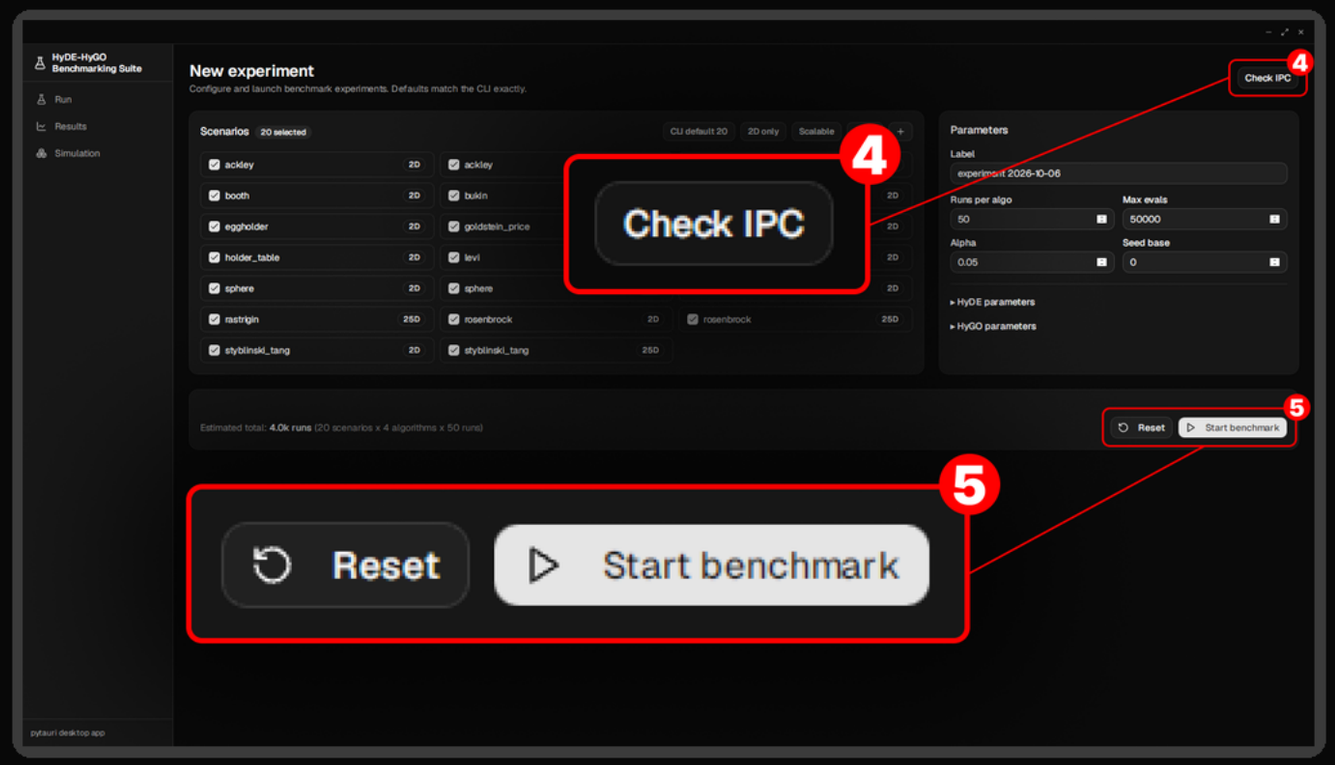
3. Set up the Parameters Panel items: Label (name of experiment), Runs per Algo (number of repetitions per algorithm), Max Evals (number of function evaluations per repetition), Alpha, and Seed Base (the initial value of the random seed), or open advanced settings in HyDE Parameters and HyGO Parameters;

4. Use Check IPC to check the connection between the application interface and the Python backend before running; and

5. Use Reset to reset the settings or Start Benchmark to start the experiment, with the number of runs shown as Estimated Total Runs.

2 annotated screenshots





MONITOR A RUNNING BENCHMARK

STEP-BY-STEP WORKFLOW / PART 1 OF 3

INSTRUCTIONS

After clicking the Start Benchmark button, users can use the Live screen to monitor the experiment they just configured, where they can:

1. View the Running Status to ensure that the benchmark process is running;
2. Click Cancel to end the benchmarking process prior to its completion;
3. View Progress to check the number of runs, number of completed scenarios, time elapsed, estimated time of arrival, and current scenario;

2 annotated screenshots

MONITOR A RUNNING BENCHMARK

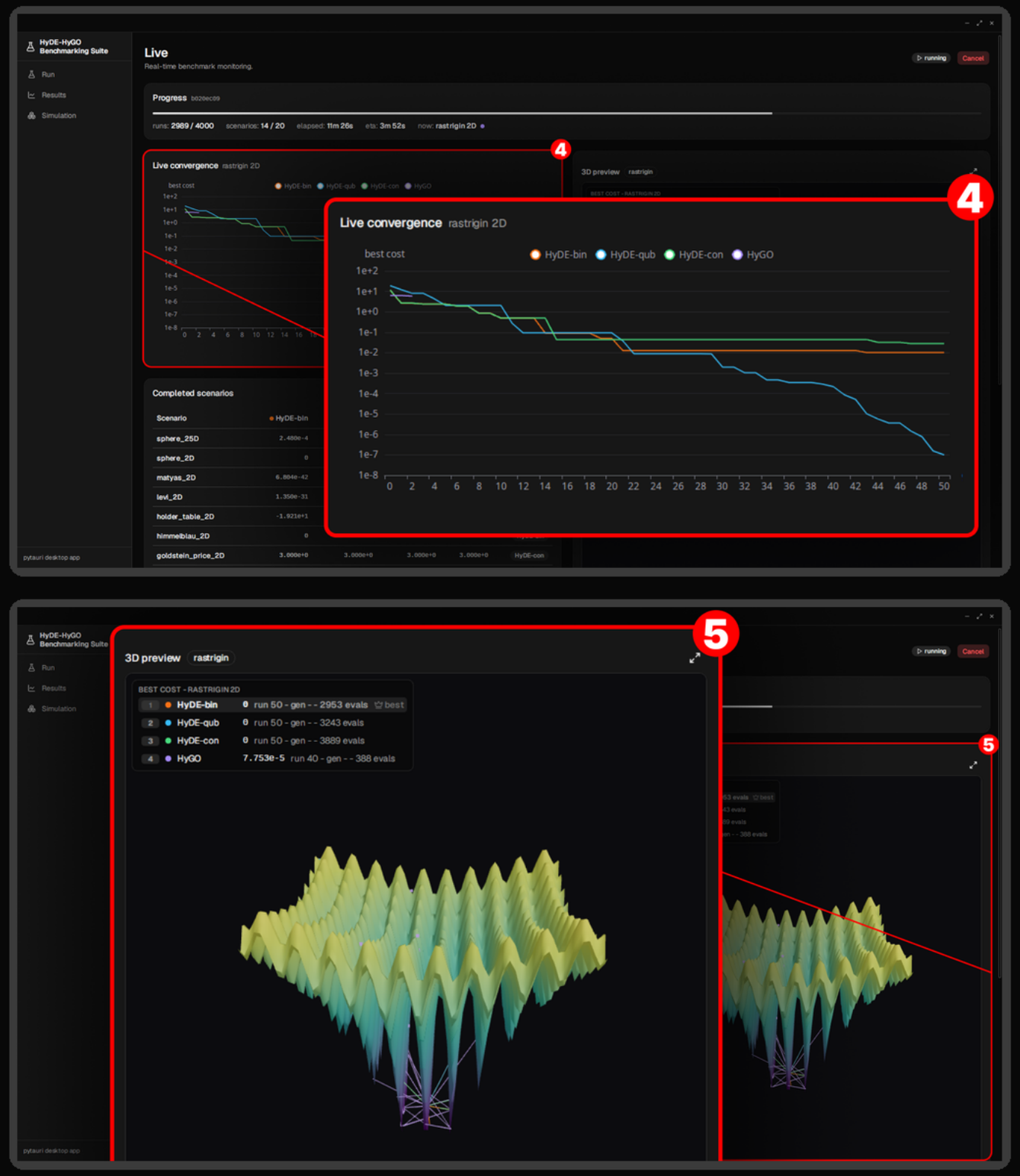
STEP-BY-STEP WORKFLOW / PART 2 OF 3

INSTRUCTIONS

4 View Live Convergence to analyze the best cost of each algorithm on the current scenario; the lower the best cost, the better;

5 View 3D Preview to analyze the surface of the current scenario;

2 annotated screenshots



MONITOR A RUNNING BENCHMARK

STEP-BY-STEP WORKFLOW / PART 3 OF 3

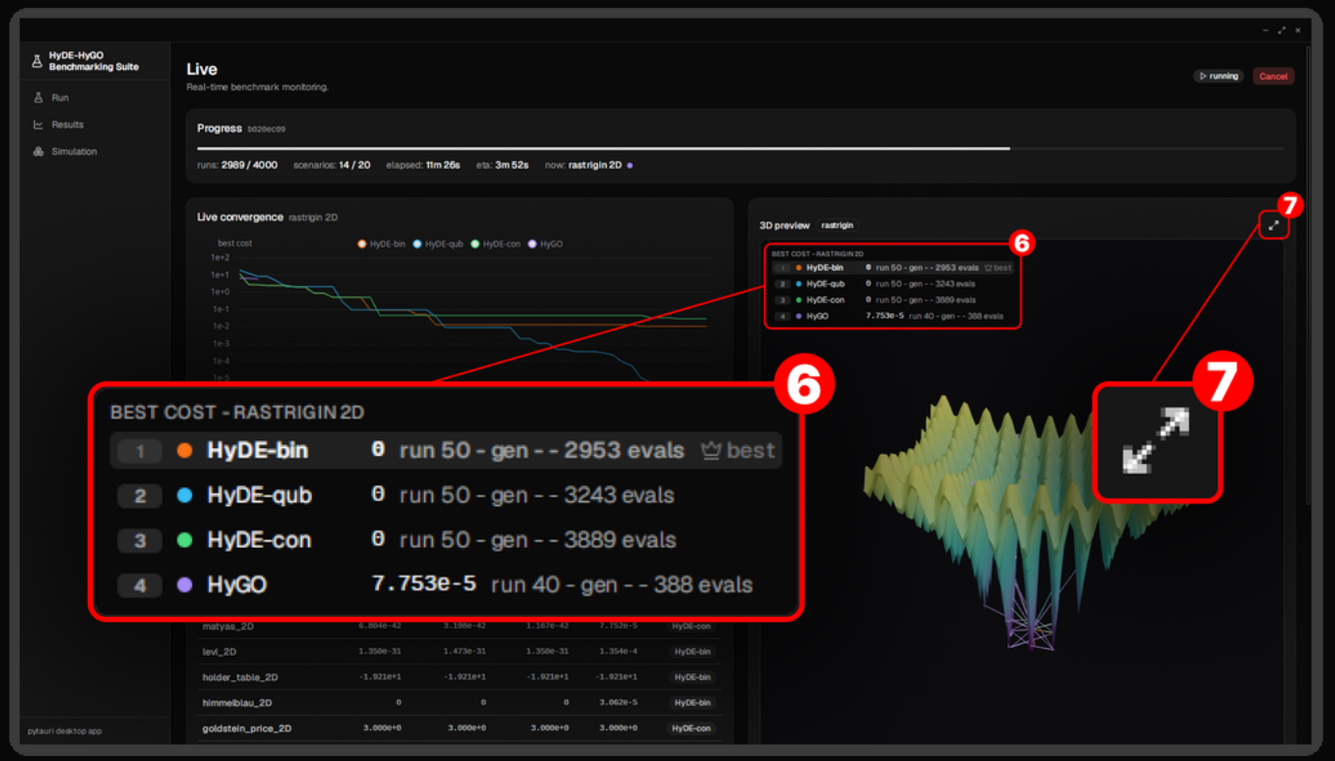
INSTRUCTIONS

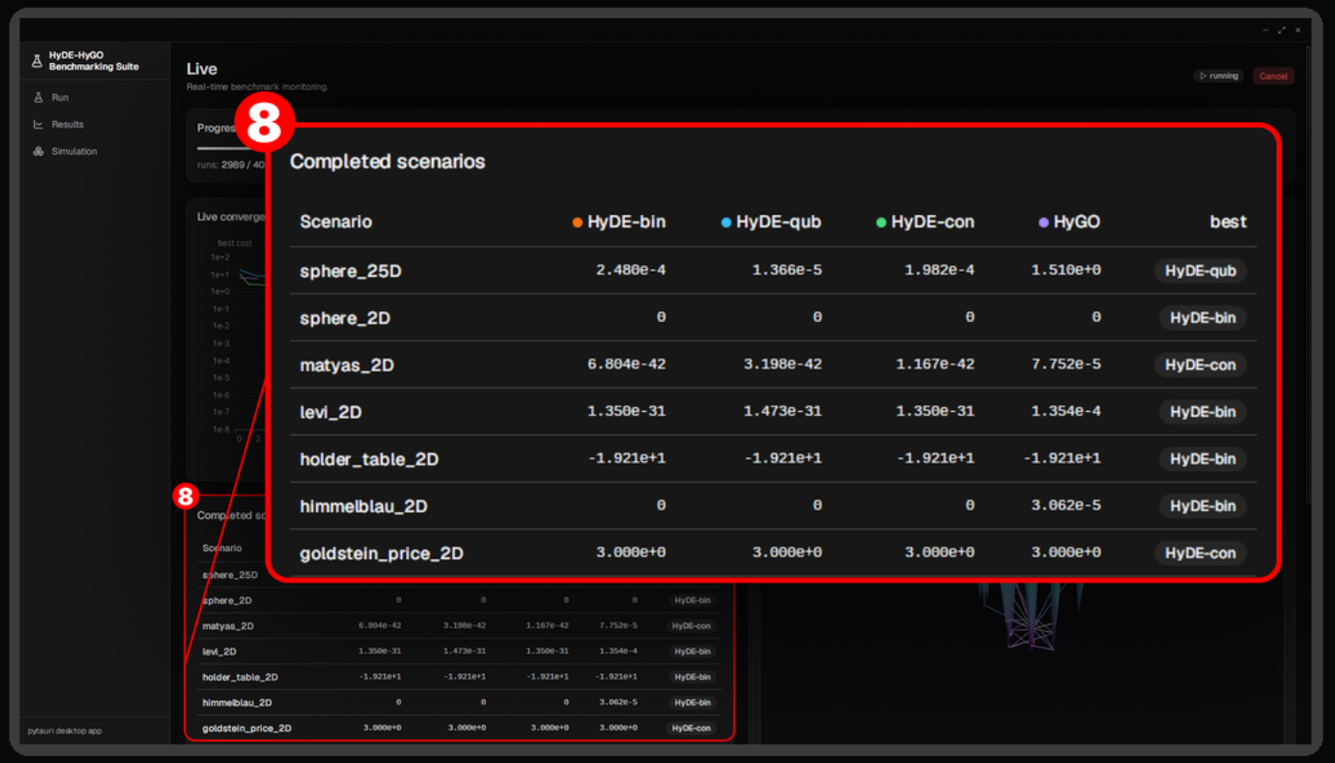
6. View Best Cost List to compare the best cost of each algorithm, number of runs, number of generations, and evaluations, with the best one being labeled as Best;

7. Click Expand to enlarge the 3D preview; and

8. View Completed Scenarios with the results of all completed scenarios for each algorithm, with the best one being labeled as Best.

2 annotated screenshots





ENLARGE THE 3D PREVIEW

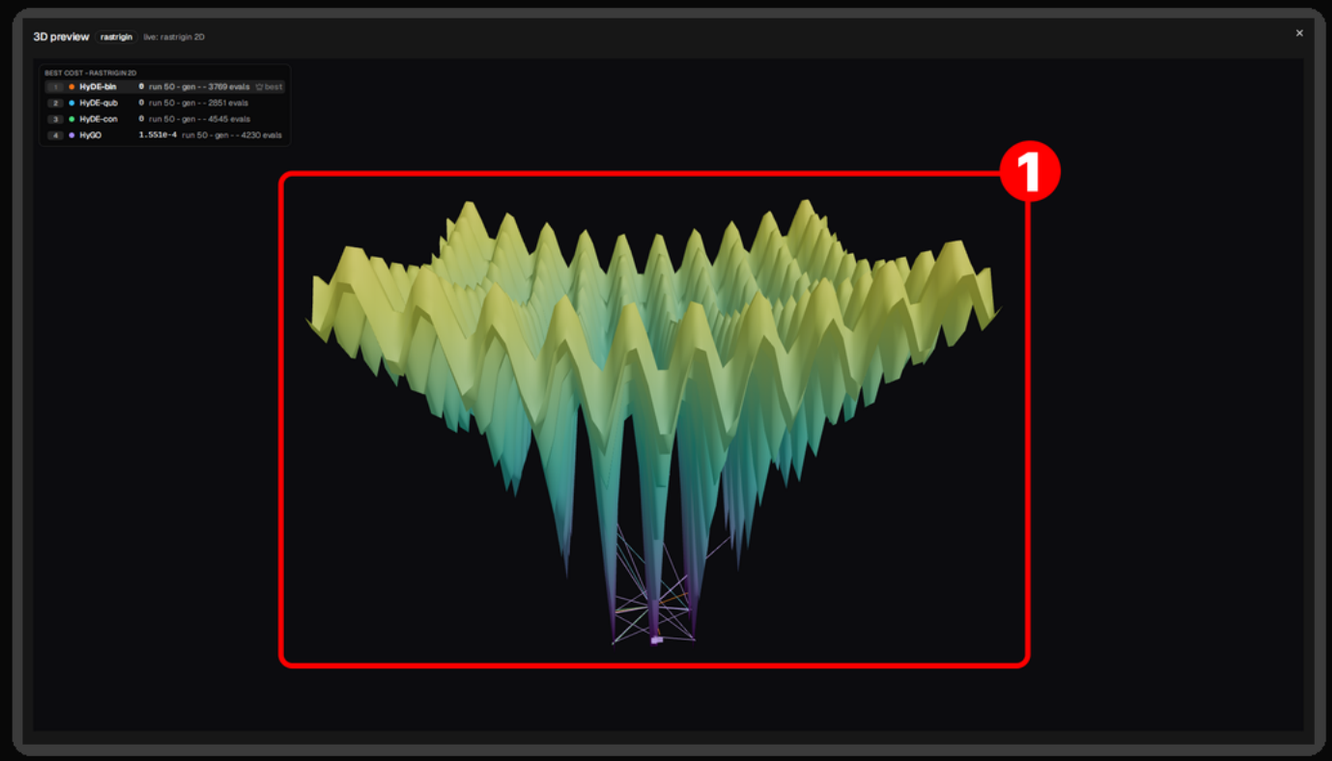
STEP-BY-STEP WORKFLOW

INSTRUCTIONS

When the Expand button is clicked, the Live 3D preview is magnified and becomes the point of focus, where users can:

1. Manipulate the orientation and position of the magnified 3D live preview of the ongoing experiment;
2. Look at the Scenario Tag to verify which test function is being viewed;
3. Go to the Best Cost List to monitor the progression of each algorithm in the scenario; and
4. Click Close to go back to the Live page.

2 annotated screenshots



REVIEW SAVED RESULTS

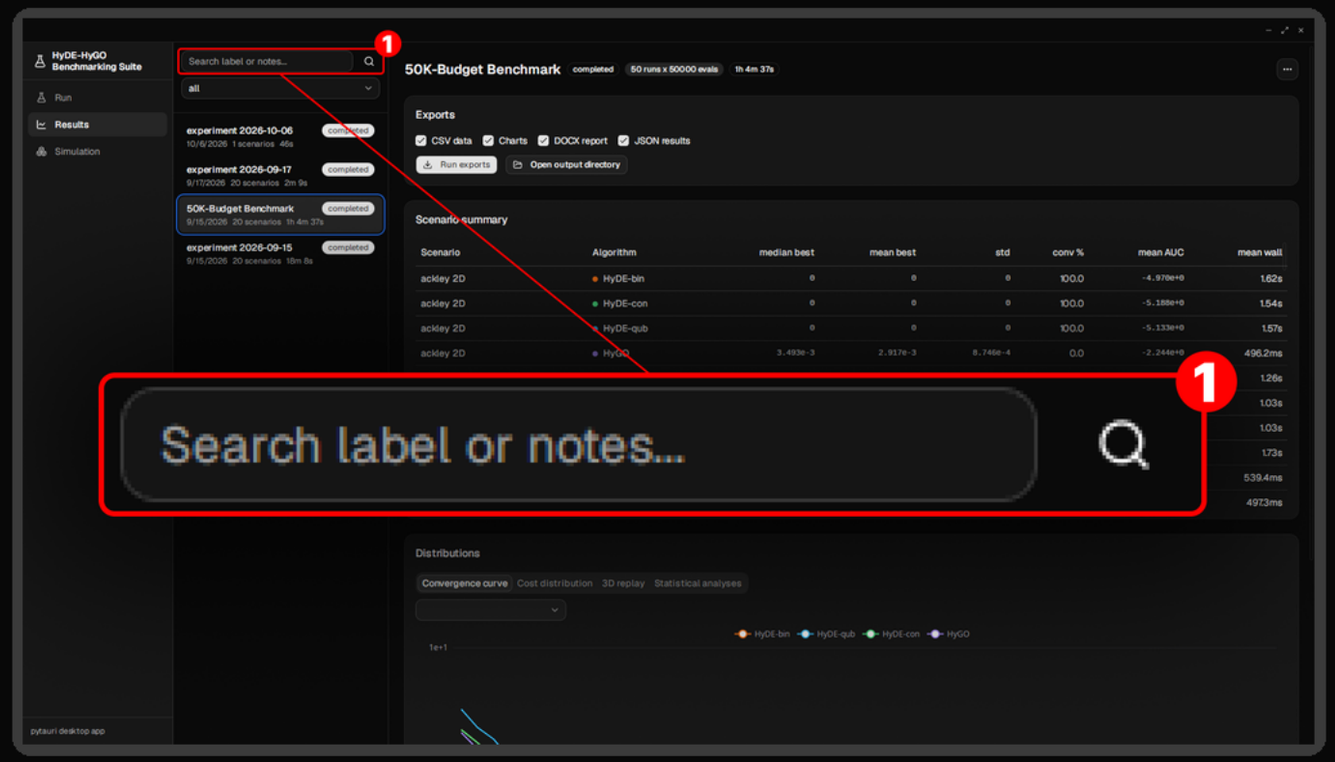
STEP-BY-STEP WORKFLOW / PART 1 OF 3

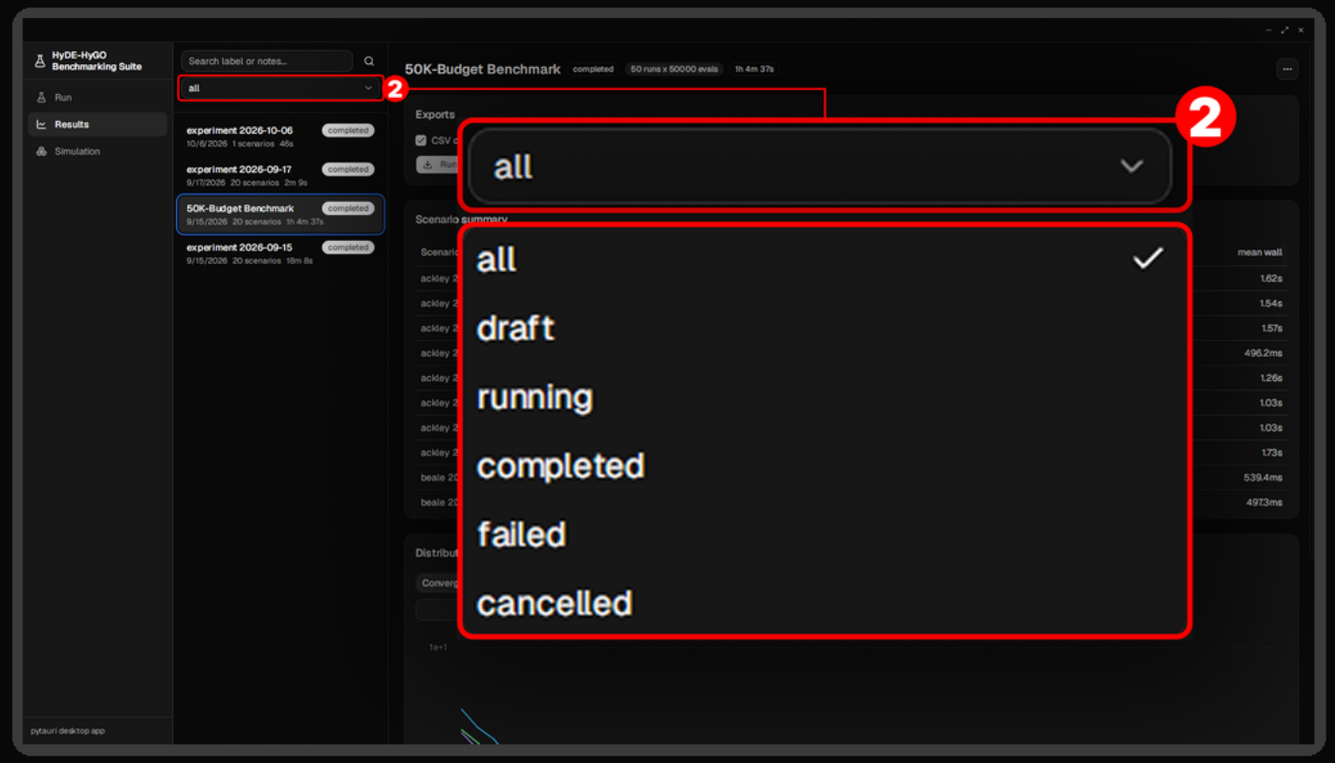
INSTRUCTIONS

When users move to the Results page, users will see the previously conducted experiments and their results, where they can:

1. Type in the Search Box to search for an experiment using its label or notes;
2. Select from the Filter Dropdown to limit the list of saved experiments (currently set to all);

2 annotated screenshots





REVIEW SAVED RESULTS

STEP-BY-STEP WORKFLOW / PART 2 OF 3

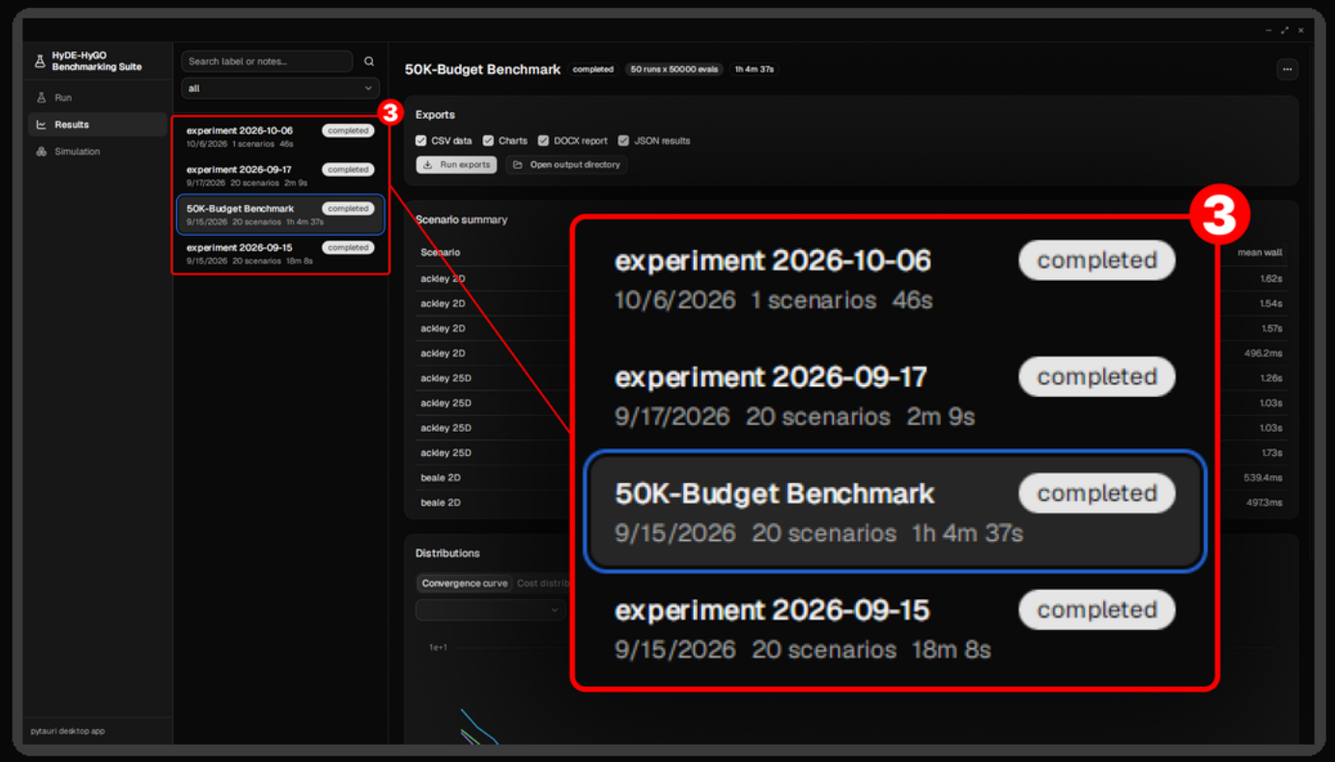
INSTRUCTIONS

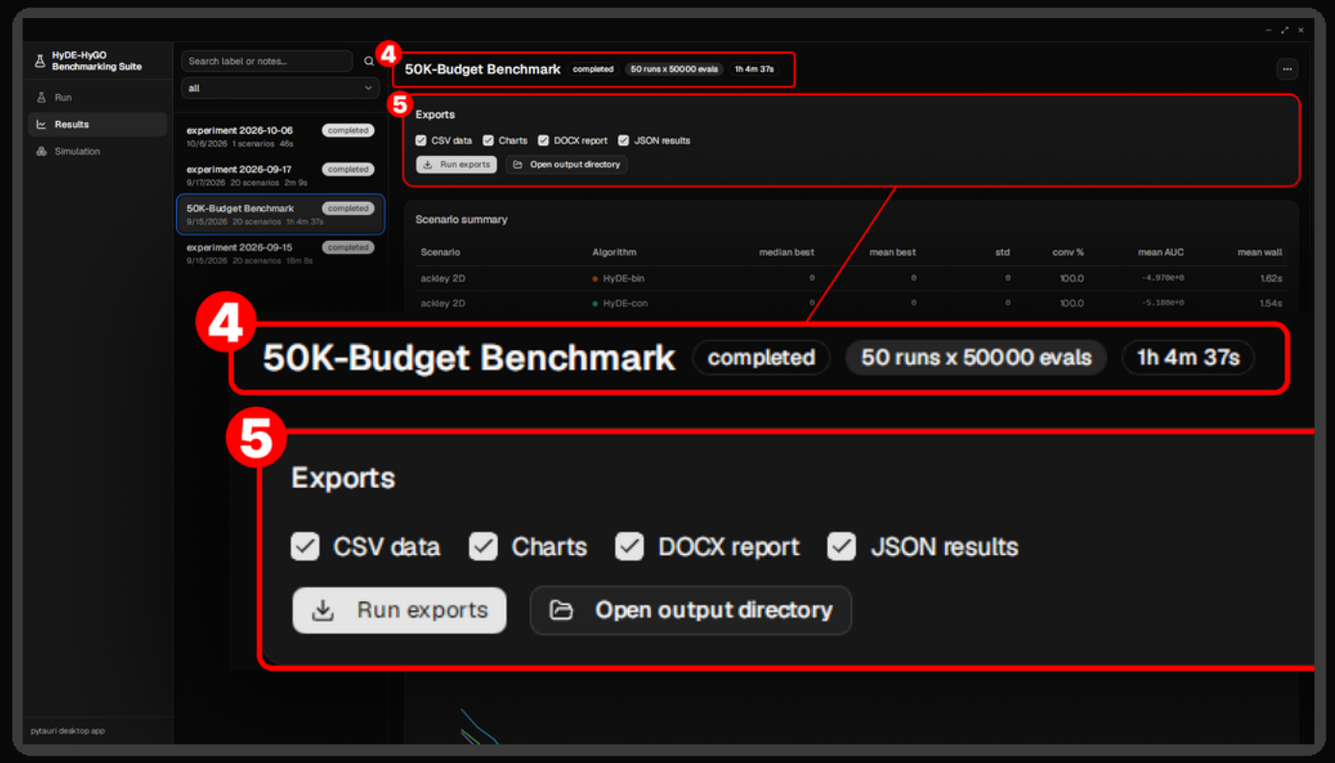
3. Select an experiment from the Experiment List to view its results;

4. See the Experiment Header that displays the experiment title, status, settings (number of runs and evaluations per run), and duration;

5. Use the Exports options to choose which file types (CSV Data, Charts, DOCX Report, and JSON Results) to export, click Run Exports to export them, or click Open Output Directory to open the output directory;

2 annotated screenshots



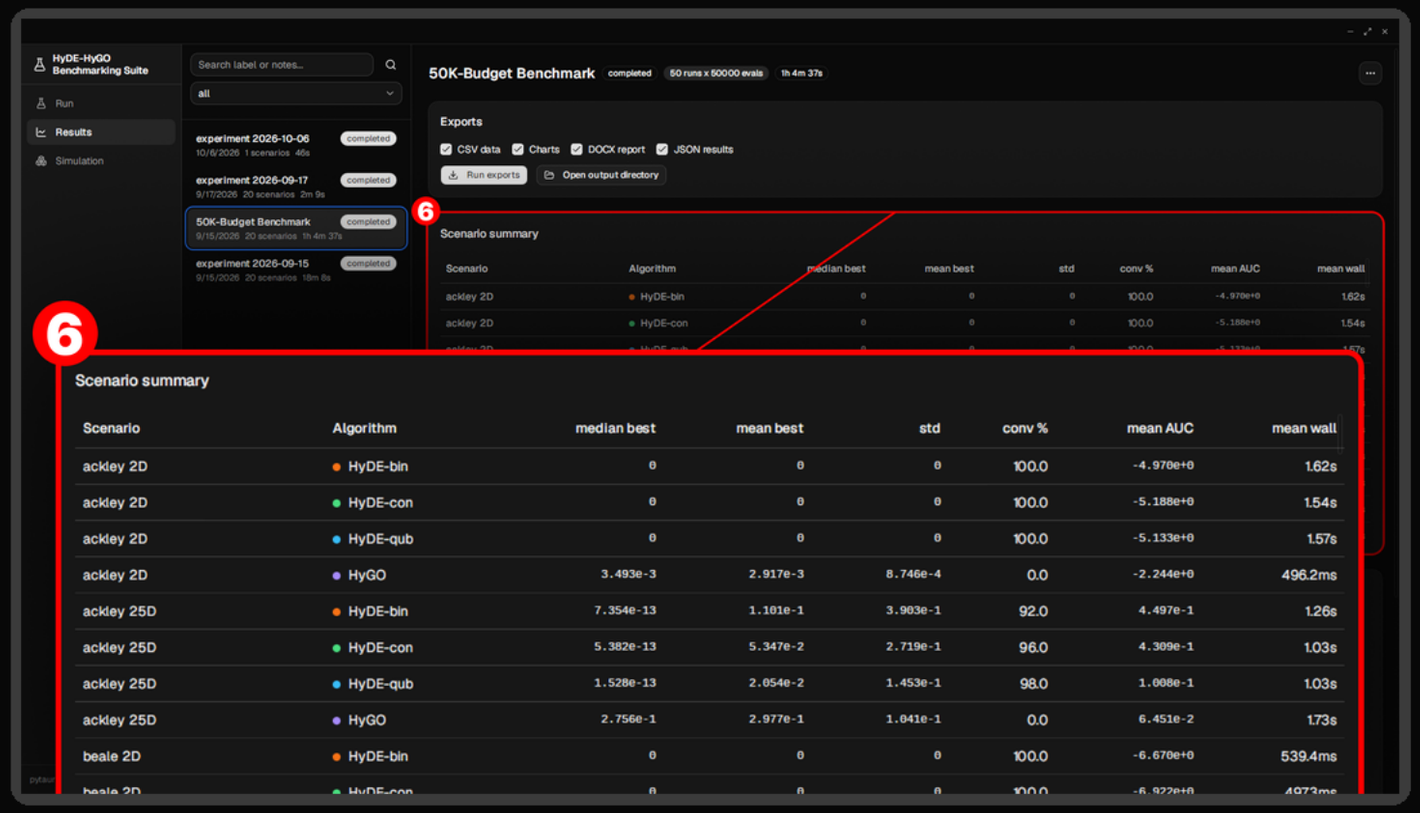


REVIEW SAVED RESULTS

STEP-BY-STEP WORKFLOW / PART 3 OF 3

INSTRUCTIONS

6 See the Scenario Summary that compares the algorithms used for each scenario in terms of best cost, standard deviation, convergence rate, and average time;



1 annotated screenshot

MANAGE AN EXPERIMENT

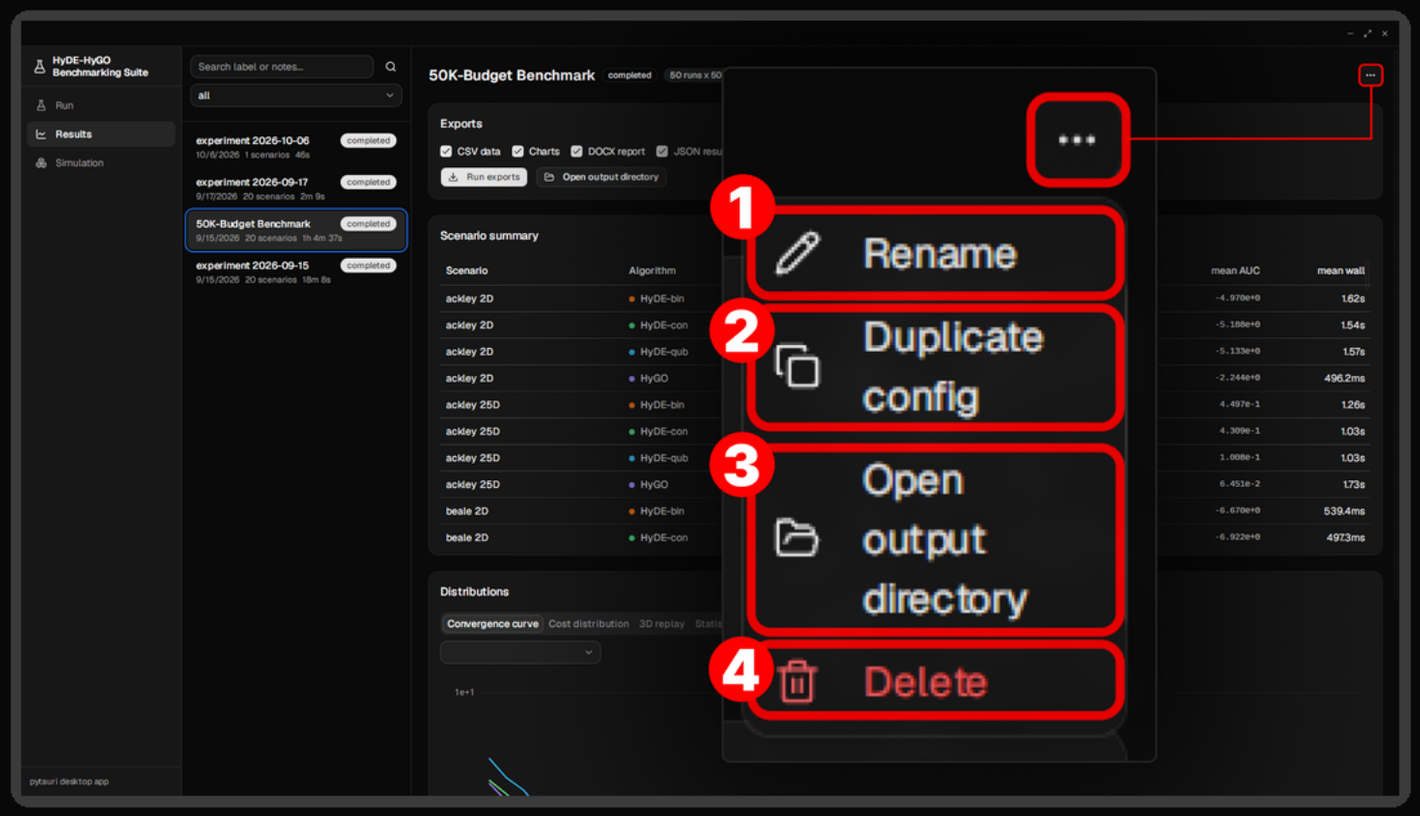
STEP-BY-STEP WORKFLOW

INSTRUCTIONS

In the Results page, when users click the (...) Menu button, they can:

1. Choose Rename to rename the experiment;
2. Choose Duplicate Config to duplicate the configuration of the experiment;
3. Choose Open Output Directory to access the directory that contains the experiment files; and
4. Choose Delete to delete the experiment.

1 annotated screenshot



NAVIGATE IN DISTRIBUTIONS SECTION TABS

STEP-BY-STEP WORKFLOW

INSTRUCTIONS

In the Results page, when users navigate to the Distributions panel, users can choose from the following tabs:

1. Click Convergence Curve to see how the best cost of each algorithm decreases across generations;
2. Click Cost Distribution to see how the final results of each algorithm are spread;
3. Click 3D Replay to replay a recorded search step by step; and
4. Click Statistical Analyses to see the tests that show whether the differences between the algorithms are significant.

1 annotated screenshot

VIEW THE CONVERGENCE CURVE

STEP-BY-STEP WORKFLOW / PART 1 OF 2

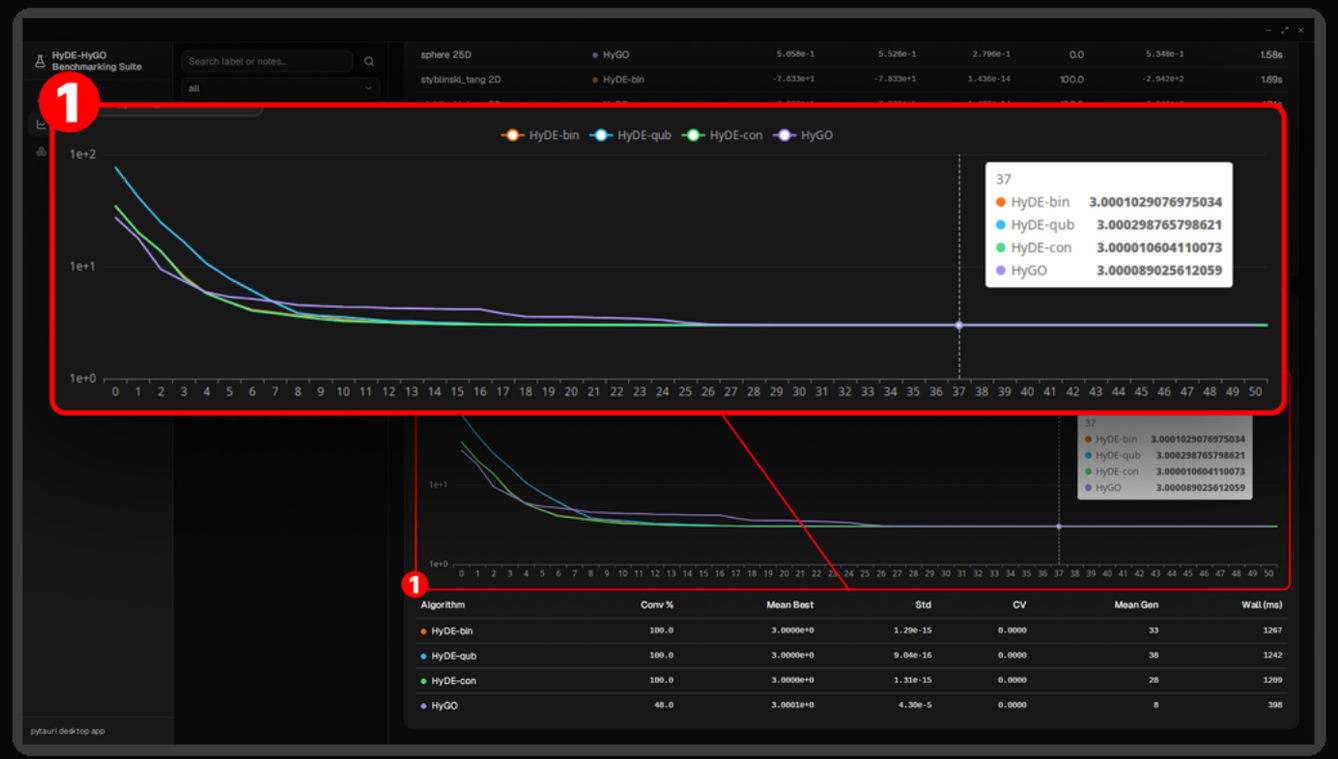
INSTRUCTIONS

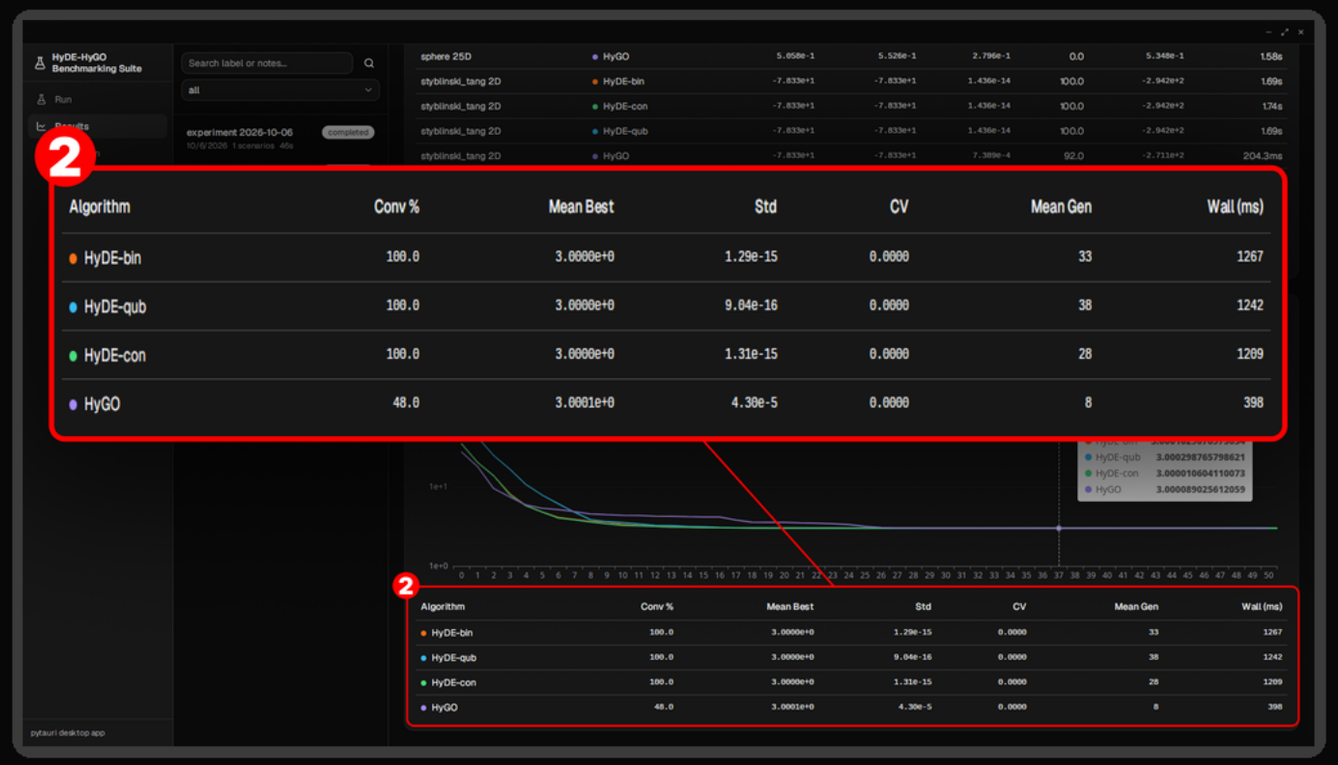
In the Results page, once users click the Convergence Curve tab, they can see how the best cost of each algorithm changes across generations, where they can:

1. Hover over the chart to see the Tooltip showing the actual best cost of each algorithm at the selected generation;

2. Read the Statistics Table below the chart to compare each algorithm's Conv %, Mean Best, Std, CV, Mean Gen, and Wall (ms); and

2 annotated screenshots



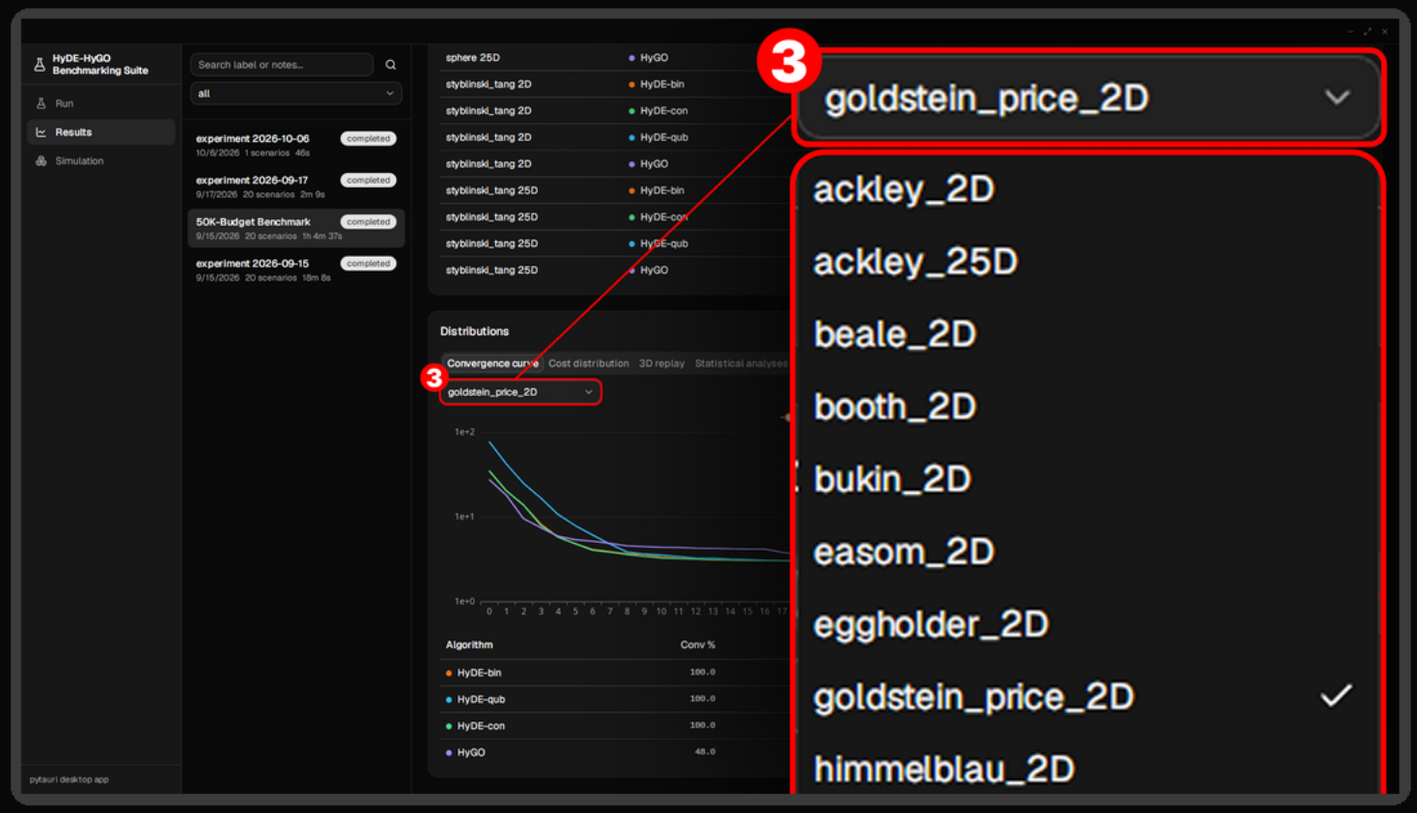


VIEW THE CONVERGENCE CURVE

STEP-BY-STEP WORKFLOW / PART 2 OF 2

INSTRUCTIONS

3 Select the required scenario using the Scenario Dropdown to view its convergence curve.



1 annotated screenshot

VIEW THE COST DISTRIBUTION

STEP-BY-STEP WORKFLOW

INSTRUCTIONS

In the Results page, once users click the Cost Distribution tab, users can see how the final results of each algorithm are spread, where they can:

1. View the Cost Distribution Chart, which shows a box plot of the final cost for each algorithm; a lower, thinner box means better results.

1 annotated screenshot

REPLAY THE SEARCH IN 3D

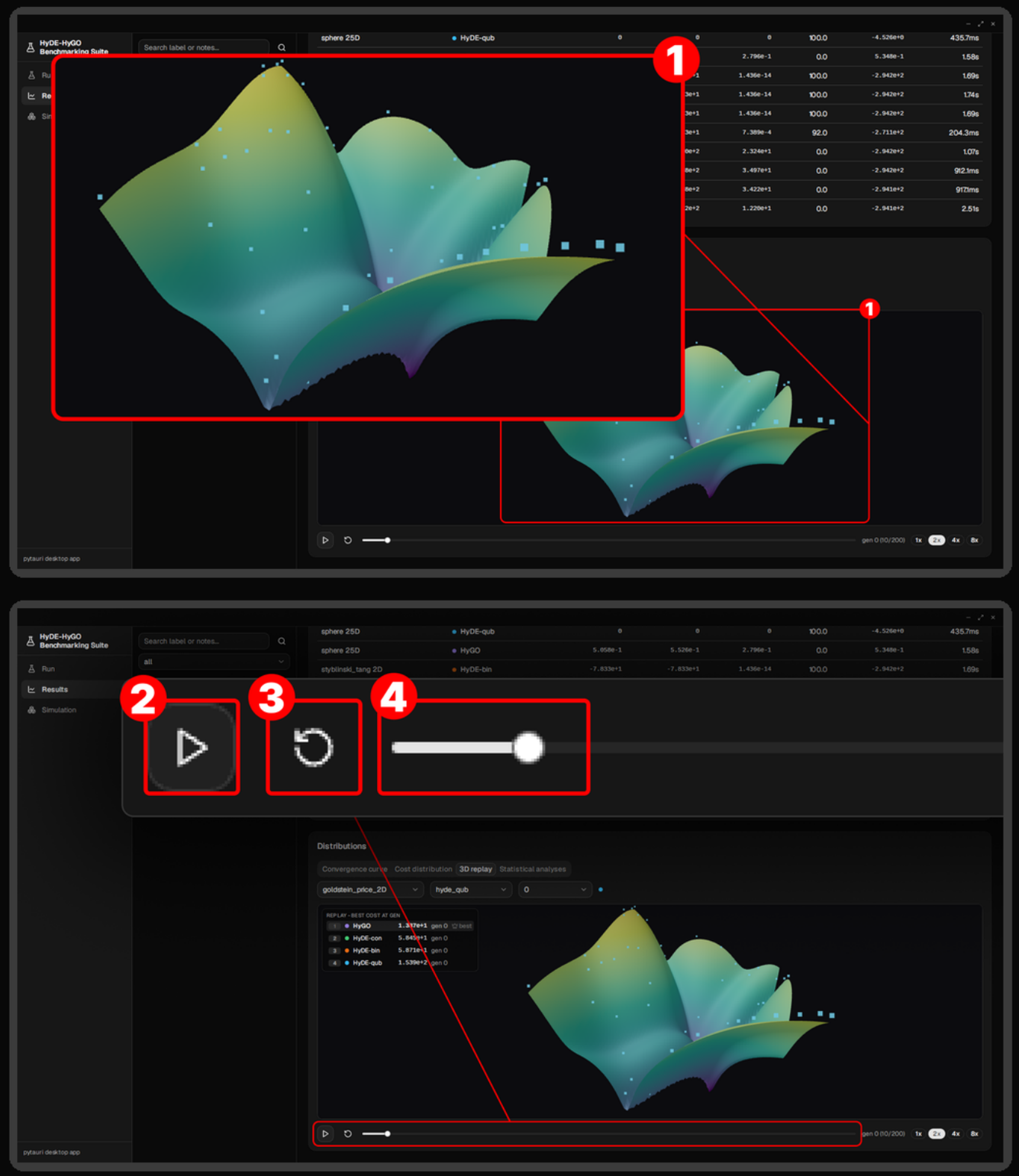
STEP-BY-STEP WORKFLOW / PART 1 OF 3

INSTRUCTIONS

In the Results page, when users click the 3D Replay tab, they can replay a recorded search step by step, where they can:

1. See the 3D Surface with the dots, which represent the candidate solutions generated by the algorithm in the current generation, and the Replay Best Cost List ranking the algorithms from best to worst for that generation;
2. Click Play to start the replay;
3. Click Reset to return the replay to its initial state;
4. Drag the Progress Slider to move to a specific generation of the replay;

2 annotated screenshots



REPLAY THE SEARCH IN 3D

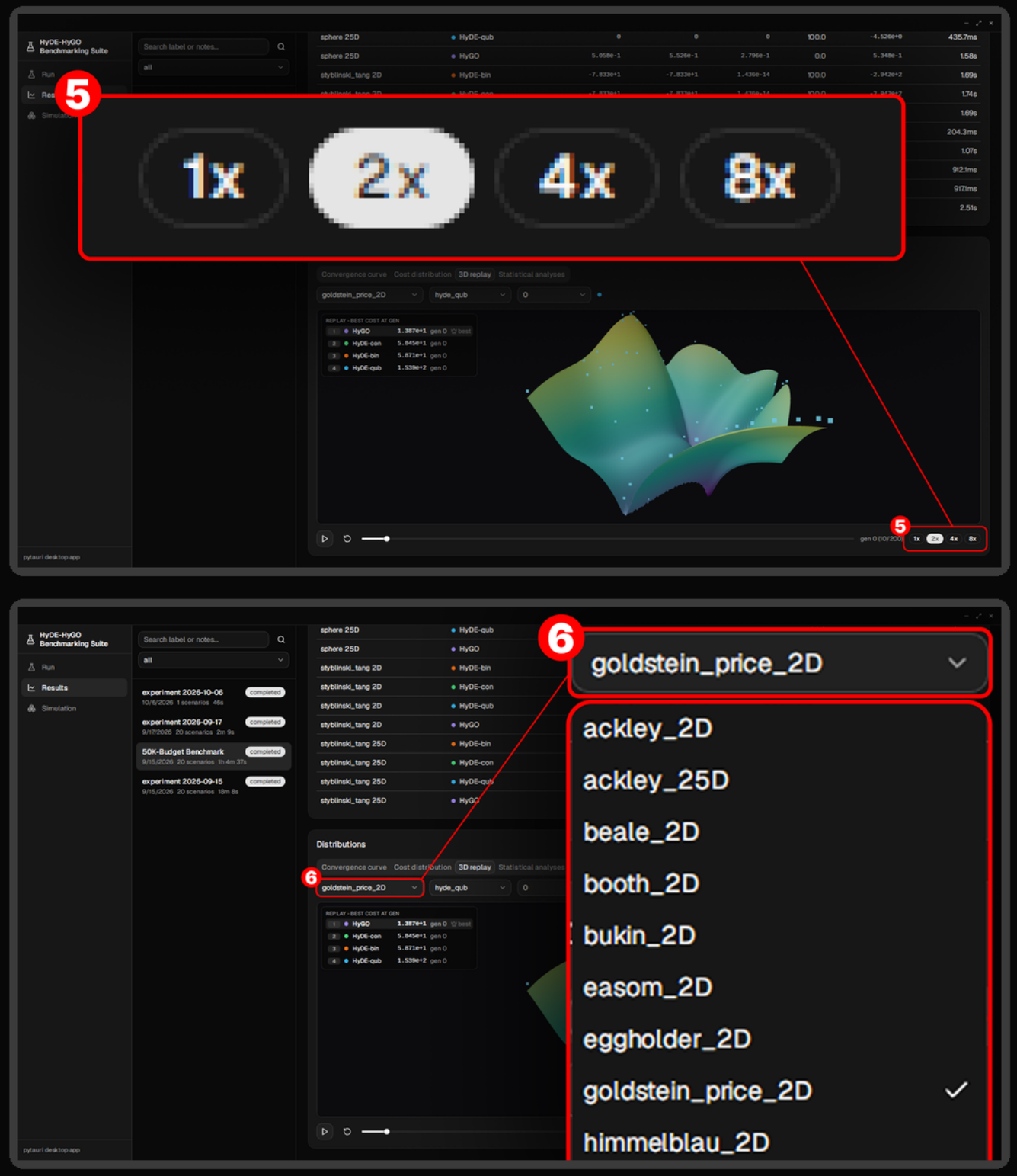
STEP-BY-STEP WORKFLOW / PART 2 OF 3

INSTRUCTIONS

5 Click Speed (1x, 2x, 4x, or 8x) to change how fast the replay runs;

6 Click the Scenario Dropdown to select the scenario to replay;

2 annotated screenshots



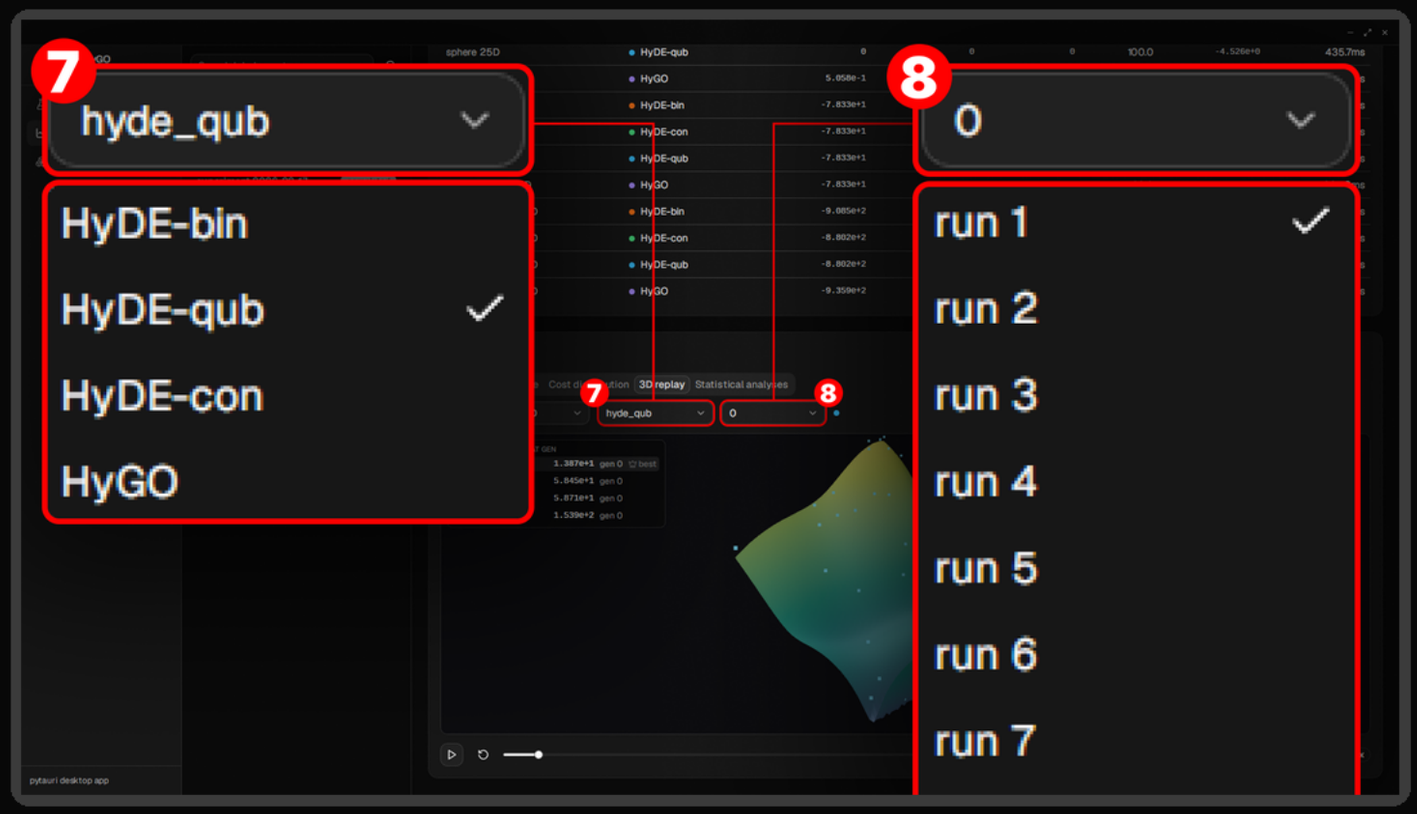
REPLAY THE SEARCH IN 3D

STEP-BY-STEP WORKFLOW / PART 3 OF 3

INSTRUCTIONS

7. Click the Algorithm Dropdown to select the algorithm to replay; and

8. Click the Run Dropdown to select the run of the chosen algorithm to replay.



1 annotated screenshot

READ THE STATISTICAL ANALYSIS

STEP-BY-STEP WORKFLOW

INSTRUCTIONS

In the Results page, when users click the Statistical Analyses tab, users can see the tests that indicate the significance of the differences between the algorithms, as well as charts of how consistently each algorithm performs, where they can:

1. Analyze the Statistical Tables to learn which algorithm wins in each pair (Rank Diff, CD, Significant, Direction) and in each scenario (H-stat, p-value, Sig., Best), where a pair is considered significant if its Rank Diff exceeds the CD; and
2. View the Result Consistency chart (coefficient of variation per benchmark at 25D) and the Normalized Degradation chart (from 2D to 25D; a lower value means the algorithm scales better).

2 annotated screenshots

ACCESS THE SIMULATION SCREEN

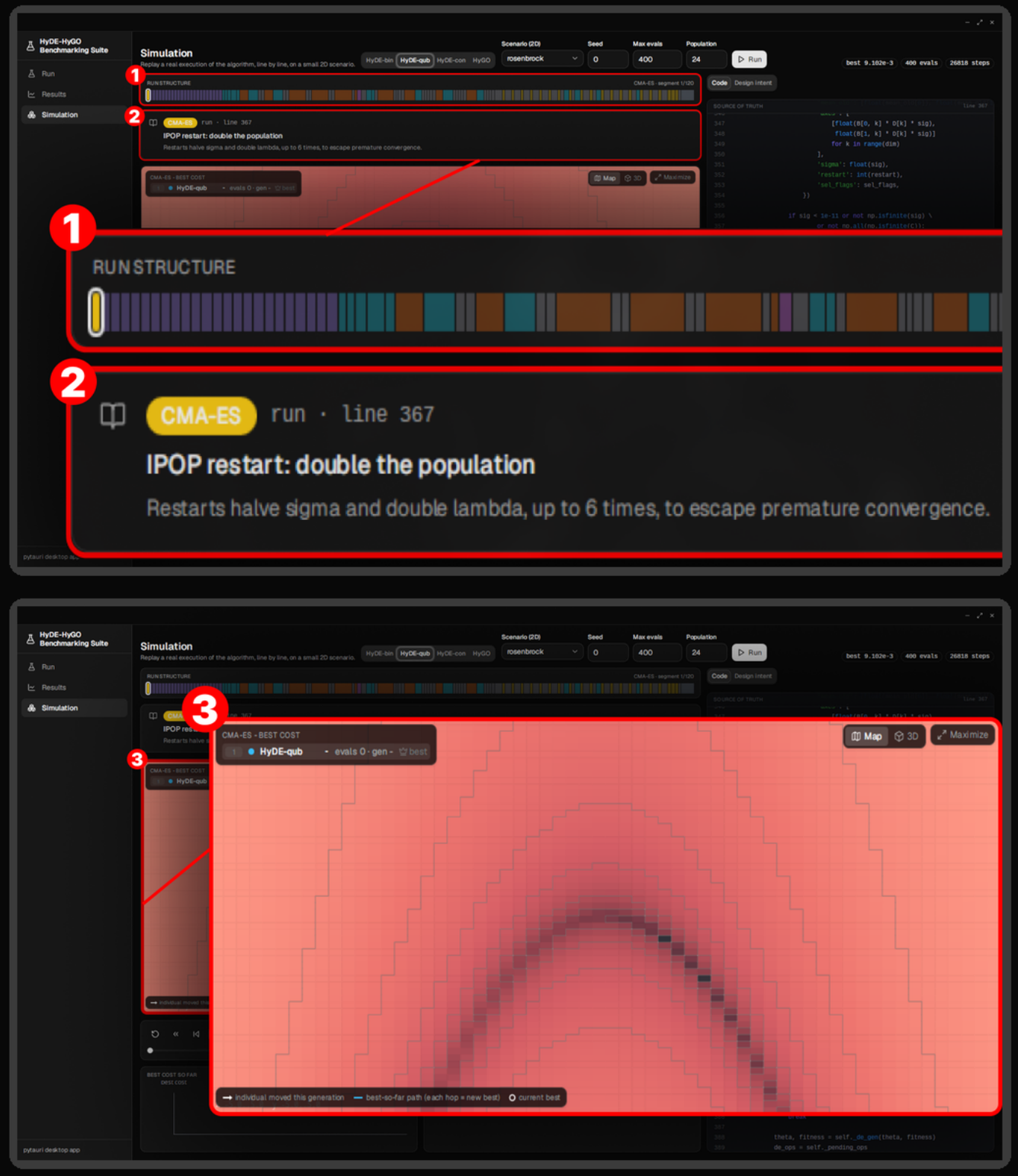
STEP-BY-STEP WORKFLOW / PART 1 OF 5

INSTRUCTIONS

In the Simulation page, when users run an algorithm, they can follow its execution line by line on a small 2D scenario, where they can:

1. See the Run Structure, a color-coded timeline of the whole run, indicating the current segment and stage;
2. See the Stage Card, which shows the current stage of the algorithm, the name of the function and the line of code being run, and an explanation of it;
3. See the Simulation Display with the search process in the selected scenario, together with the Best Cost Label showing the current stage, best cost, evaluation count, and generation;

2 annotated screenshots



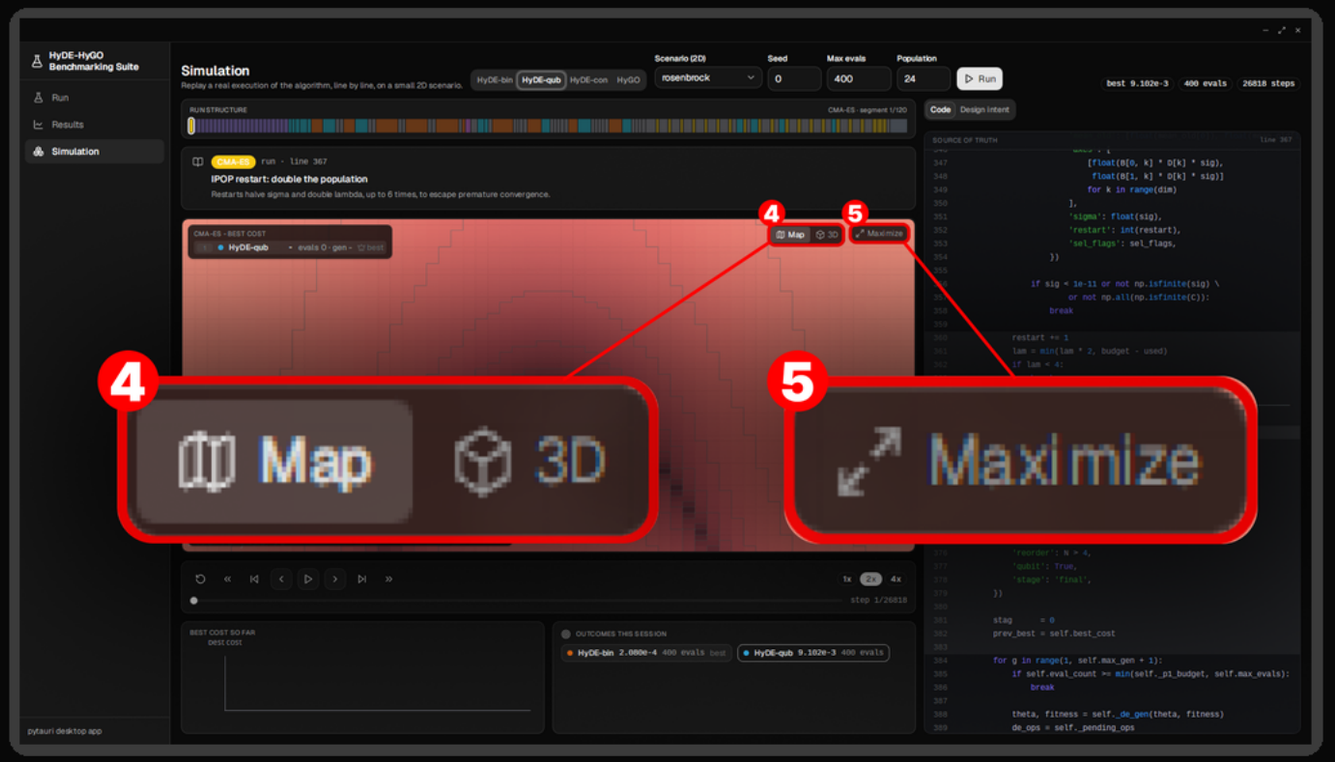
ACCESS THE SIMULATION SCREEN

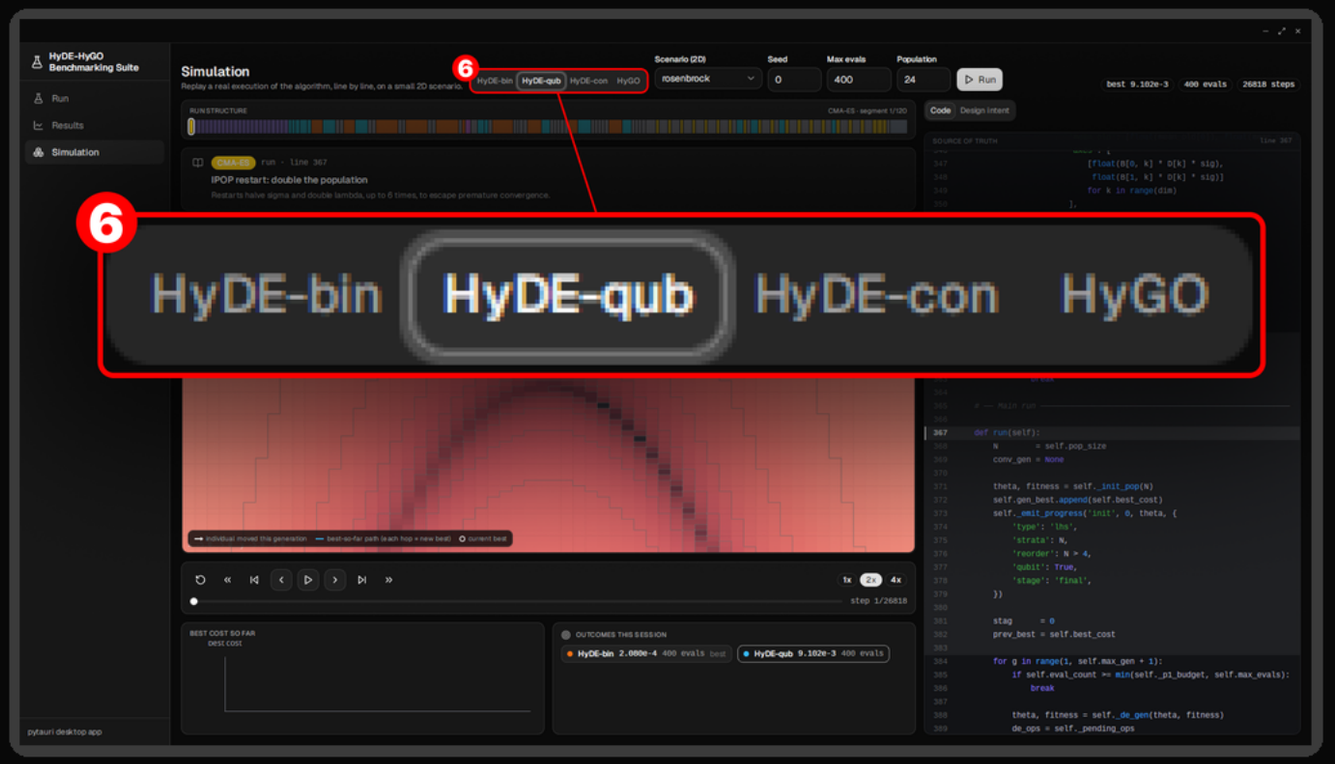
STEP-BY-STEP WORKFLOW / PART 2 OF 5

INSTRUCTIONS

4. Click the Map or 3D buttons to switch between the top-down map and the 3D surface;
5. Click the Maximize button to enlarge the simulation display;
6. Click the Algorithm Tabs (HyDE-bin, HyDE-qub, HyDE-con, or HyGO) to select the algorithm you want to simulate;

2 annotated screenshots





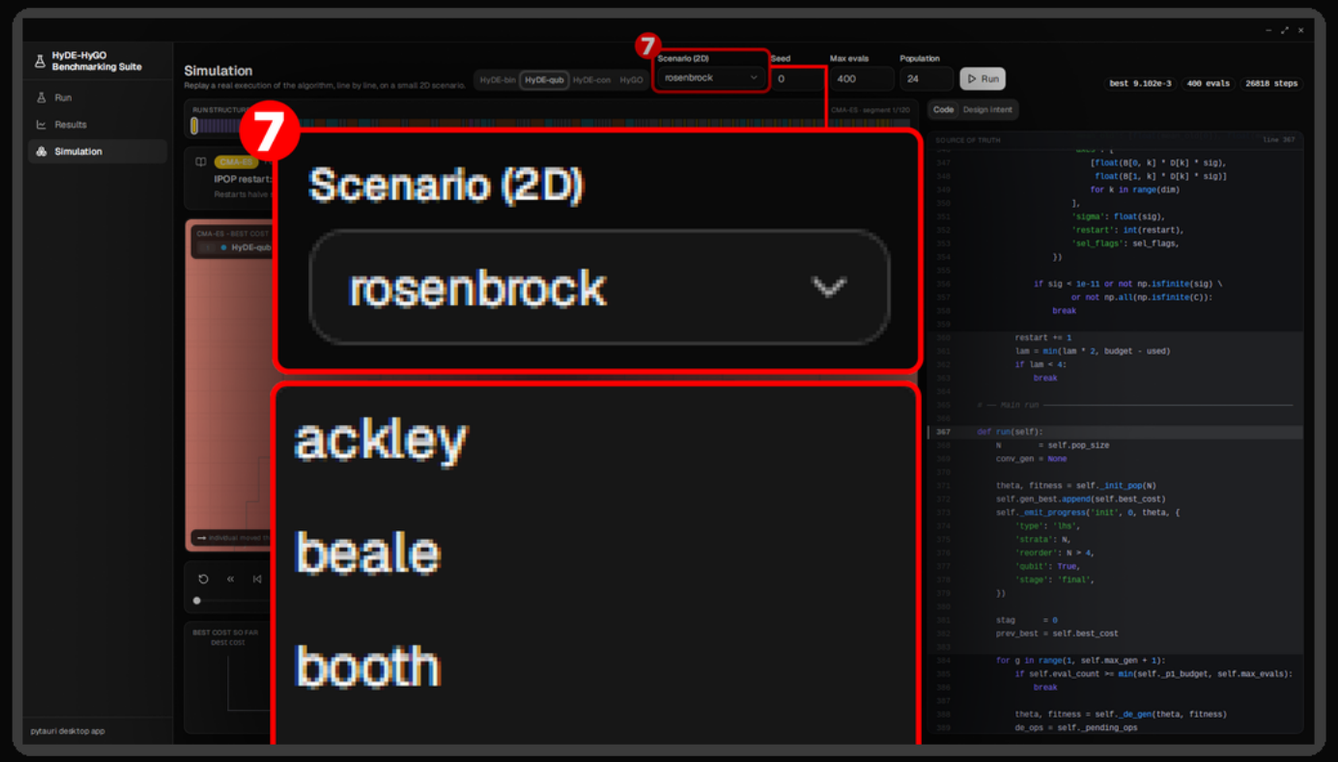
ACCESS THE SIMULATION SCREEN

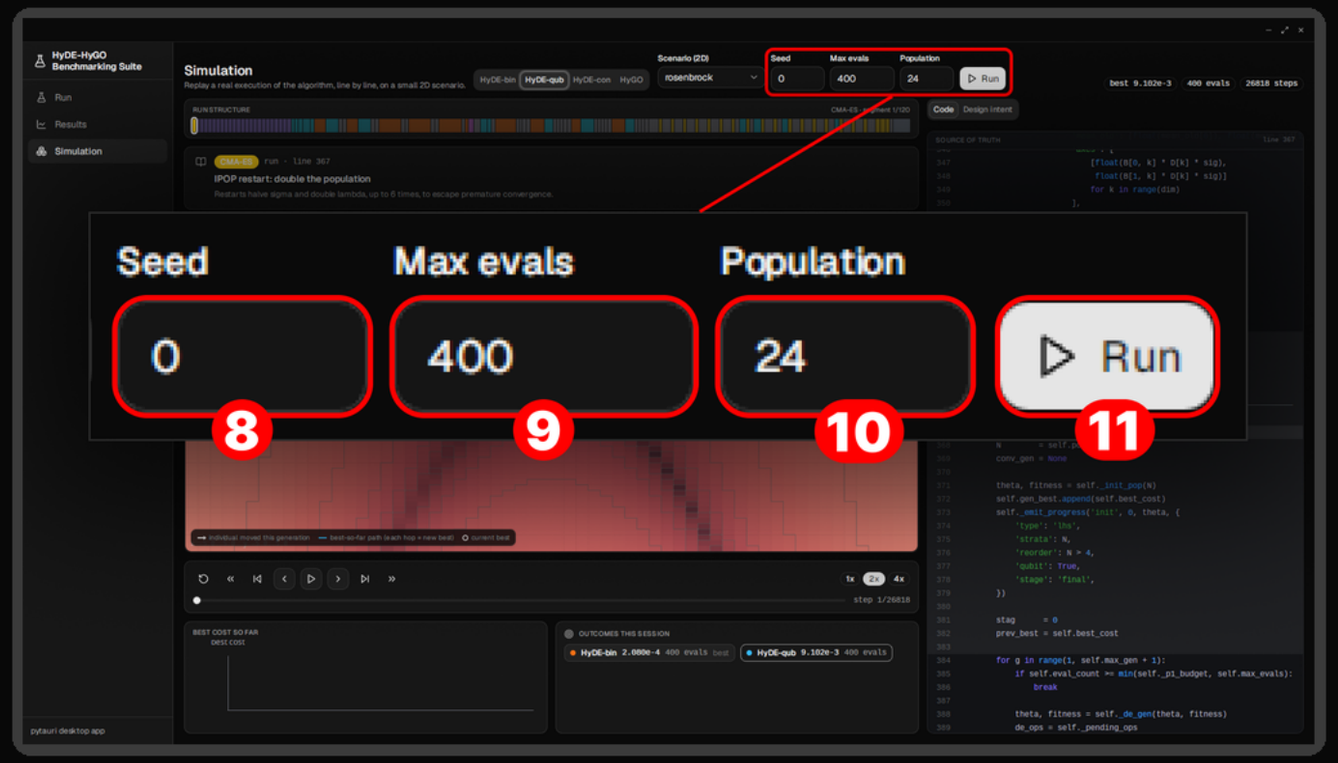
STEP-BY-STEP WORKFLOW / PART 3 OF 5

INSTRUCTIONS

7. Click the Scenario dropdown to select a 2D test function for the simulation;
8. Type the Seed to set the initial value of the random seed;
9. Type the Max Evals to set the evaluation limit;
10. Type the Population to set the number of candidates;
11. Click Run to execute the algorithm with the specified parameters and get the Run Summary with the best cost, evaluations, and total steps;

2 annotated screenshots





ACCESS THE SIMULATION SCREEN

STEP-BY-STEP WORKFLOW / PART 4 OF 5

INSTRUCTIONS

12 Control the simulation using the Playback Controls (Play, Pause, Step Forward/Backward, and Restart) and choose one of three speeds (1x, 2x, or 4x);

13 See the Source of Truth, the actual code of the algorithm with the currently running line highlighted;

2 annotated screenshots

ACCESS THE SIMULATION SCREEN

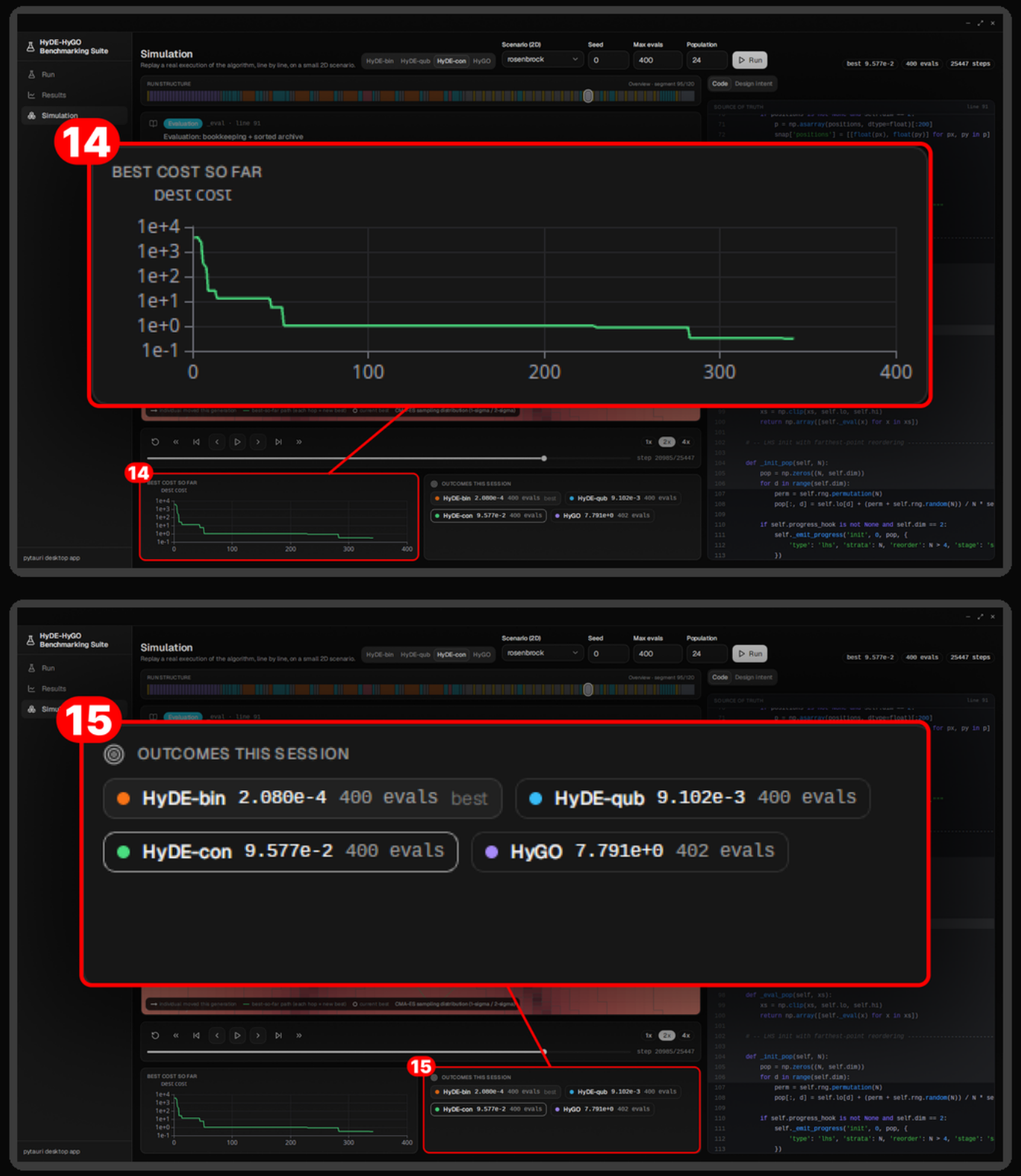
STEP-BY-STEP WORKFLOW / PART 5 OF 5

INSTRUCTIONS

14 See Best Cost So Far, the chart showing the best cost achieved in the course of the simulation; and

15 View Outcomes This Session to compare the best costs and evaluations of the simulated algorithms in the current session.

2 annotated screenshots



SWITCH AND ENLARGE THE SIMULATION DISPLAY

STEP-BY-STEP WORKFLOW

INSTRUCTIONS

In the Simulation page, users can change or maximize the view to examine the search more closely, where they can:

1. See the Maximized 3D Surface, the enlarged 3D view with the playback controls beneath it;
2. Click Minimize to restore the original size, or click the Map or 3D button to switch between the top-down map and the 3D surface; and
3. See the Maximized Map, the enlarged top-down map with the playback controls beneath it.

2 annotated screenshots